

three comes with its own strengths and weaknesses and are best at their own game. Let us have a look at how they fare against each other.

Arduino

Advantages:

- ◆ Cheap
- ◆ Number of spin offs and alternatives
- ◆ Low power consumption
- ◆ Insanely popular
- ◆ Interfaces with almost anything

Disadvantages:

- ◆ Command line interface can be an entry barrier.
- ◆ Cannot handle complex tasks with multiple processes.
- ◆ Less preferred for display oriented tasks.

Good for:

- ◆ Beginners
- ◆ Single purpose projects, e.g a video doorbell
- ◆ Interacting with objects in the real world, e.g door locks, blinds, washing machine
- ◆ Low power systems that need to run for longer time or on battery, e.g motion sensing switches

If you are looking for a plain and simple control panel for, say, your home lighting then Arduino is your thing. Unless you want that panel to interface with the internet, provide you the energy usage efficiency, and be fully automated then look somewhere else.

Raspberry Pi

Advantages:

- ◆ Basically a 35\$ Linux computer.
- ◆ Can be connected to a TV or monitor via HDMI and can be used with a keyboard and mouse via USB.
- ◆ Entire system can be changed by swapping SD card since OS runs from it.
- ◆ Can be online via ethernet.

Disadvantages

- ◆ Doesn't have wide interfacing capabilities
- ◆ Knowledge of Linux is required

Good for:

- ◆ Projects with GUI or Internet.